

Math 118 : Fourier Analysis, Wavelets and Signal Processing

Prof Jon Wilkening

office: 1051 Evans

OH: Tues 1:30 - 2:30

Wed 2:30 - 3:30

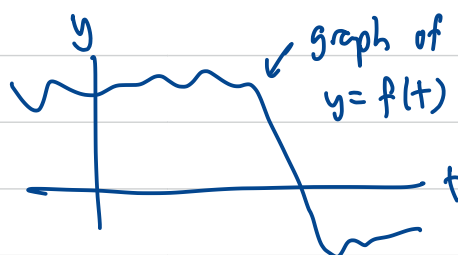
(discuss syllabus, grading, etc.)

Introduction

what is signal processing?

In this course, a "signal" is a function $f(t)$ (or $f(x)$)

$f(t)$ = value at time t



Note: we allow $f(t)$ (but not t) to be complex valued.

"Signal processing" means extracting "useful information" from the signal

Mathematical issues

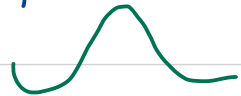
- define "useful information" in the language of operators and transforms
- develop tools to study these operators

practical issues

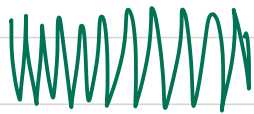
- study discrete approximations of the signal
- design fast numerical methods
- prove that they work
- implement them efficiently

examp: Radio

1. amplitude modulation (AM)



audio signal (20-20000 Hz)



carrier signal (520-1610 kHz)



2. frequency modulation (FM)

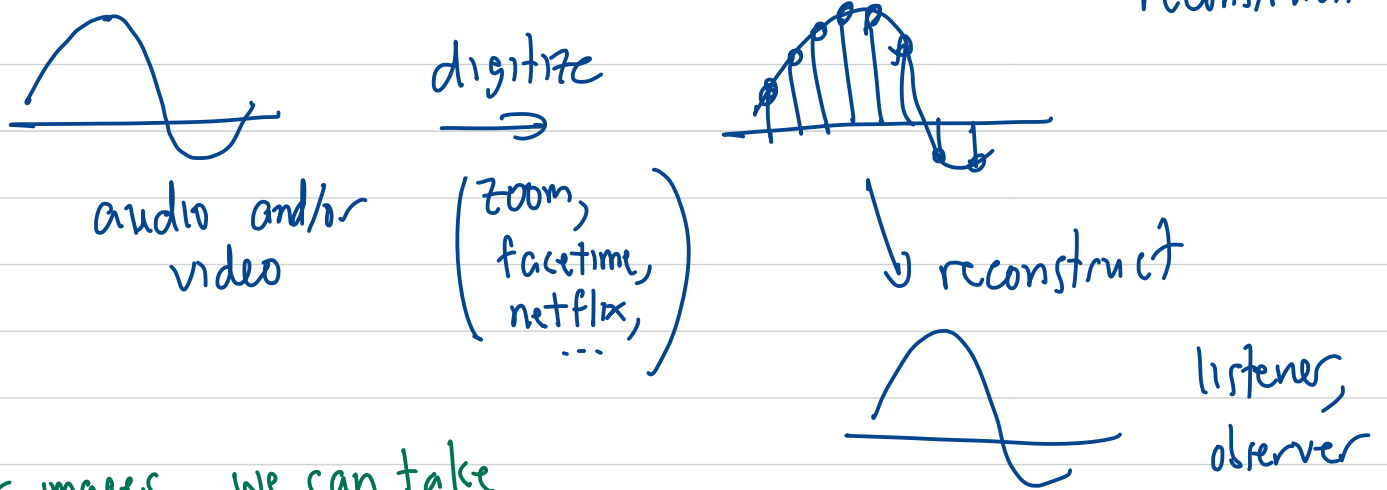
carrier signal (87.5-108.0 MHz)



frequency
varies

We need algorithms to amplify the carrier signal and extract the audio signal from it.

example: (digital) sampling, coding, compression,
reconstruction



for images, we can take
advantage of 2D spatial relationships of the data

for movies, the data can be compressed for locality
in both space and time.

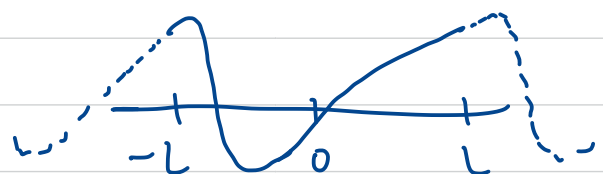
This course covers mathematical issues associated with
signal processing.

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Fourier analysis

methods of decomposing a given signal into a
linear superposition of periodic components (sines and
cosines)

Fourier series on $[-L, L]$



If f is periodic with period $2L$, then

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$$

Fourier coefficients: $\left\{ \begin{array}{l} a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx \\ a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \end{array} \right.$

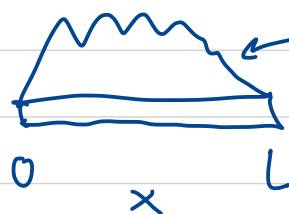
They carry "all the information" about the signal f

- If $f(x)$ is smooth (has many continuous derivatives) and periodic, the coefficients decay rapidly as $n \rightarrow \infty$ (efficient representation)

- There are fast algorithms for computing the a_n, b_n from sampled values of $f(x)$.

Application: solving the heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$



$$u(x,0) = g(x) \quad (\text{initial condition})$$

$$u(0,t) = u(L,t) = 0 \quad (\text{Boundary conditions})$$

Separation of variables (math 54) gives

$$u(x,t) = \sum_{n=1}^{\infty} b_n e^{-\alpha n^2 t / L^2} \sin\left(\frac{n\pi x}{L}\right)$$

$$g(x) = u(x,0) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \leftarrow \text{initial condition}$$

$$\rightarrow b_n = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Note that "high frequency" modes $b_n e^{-\alpha n^2 t / L^2}$ with $\frac{\alpha n^2}{L^2}$ large decay very fast as t increases.

$\rightarrow$ don't need to store many modes to represent $u(x,t)$ accurately at positive times.

Spectral methods : truncate the spatial Fourier series,
evolve the Fourier coefficients numerically in time

Complex notation : $e^{i\theta} = \cos \theta + i \sin \theta$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{n\pi x}{L}}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i \frac{n\pi x}{L}} dx$$

$n \in \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ integers

If f is real-valued

$$c_0 = a_0$$

$$\left. \begin{aligned} c_n &= \frac{1}{2}(a_n - ib_n) \\ c_{-n} &= \frac{1}{2}(a_n + ib_n) = \bar{c}_n \end{aligned} \right\} n=1, 2, 3, \dots$$

reason: substitute $e^{-i \frac{n\pi x}{L}} = \cos \frac{n\pi x}{L} - i \sin \frac{n\pi x}{L}$

how the terms match up ($n \geq 1$)

$$\left[\begin{array}{c} c_n e^{inx} + c_{-n} e^{-inx} \\ \uparrow \qquad \qquad \uparrow \\ \cos nx \pm i \sin nx \end{array} \right] = \underbrace{(c_n + c_{-n})}_{a_n} \cos nx + \underbrace{(ic_n - ic_{-n})}_{b_n} \sin nx$$

One can often non-dimensionalize the problem so that $L = \pi$.

simpler basis functions: $\varphi_n(x) = e^{inx}$, $n \in \mathbb{Z}$

What if f is defined for all $-\infty < x < \infty$ but is not periodic?

Fourier transform pair: $\hat{f}(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\lambda x} dx$

(plays the role of the coefficients c_n)

reconstruction: $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\lambda) e^{i\lambda x} d\lambda$

$\uparrow$ $\uparrow$ note $d\lambda$, not dx

still represents $f(x)$ as a superposition of periodic functions

The Fourier transform has many applications beyond signal processing. (e.g. quantum mechanics)

What are wavelets? Alternatives to sinus and cosines

retain:

- orthonormal basis
- good scaling properties

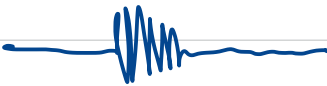
(they are all built from a single function up to scaling and translation)

give up smoothness and periodicity in favor of localization

example where Fourier is better:

height of tides. Makes sense to decompose the signal in terms of periodic functions to observe the effects of daily rotation of the Earth, monthly moon orbit, yearly earth orbit, etc.

example where Wavelets are better:

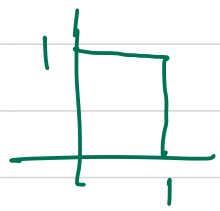
seismic signal 

most of the time nothing happens.

but need a lot of Fourier coefficients to model the signal

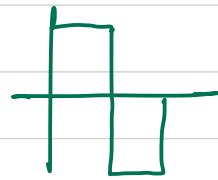
by contrast, most coefficients in the wavelet expansion are zero and don't have to be stored

Simplist case: Haar wavelet on $[0,1]$



$$\phi(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

scaling function



$$\psi(x) = \begin{cases} 1 & 0 \leq x < \frac{1}{2} \\ -1 & \frac{1}{2} \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

wavelet

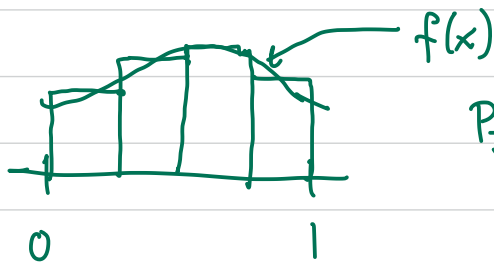
$$f(x) = a_0 \phi(x) + \sum_{j=0}^{\infty} \sum_{k=0}^{2^j-1} b_{jk} \psi\left(2^j\left(x - \frac{k}{2^j}\right)\right)$$

↑ scaling ↑ translation

$$a_0 = \int_0^1 f(x) \phi(x) dx, \quad b_{jk} = 2^j \int_0^1 f(x) \psi\left(2^j\left(x - \frac{k}{2^j}\right)\right) dx$$

If we truncate at level J ($P_J f(x) = a_0 \phi(x) + \sum_{j=0}^{J-1} \dots$)

We get a step function approximation to $f(x)$



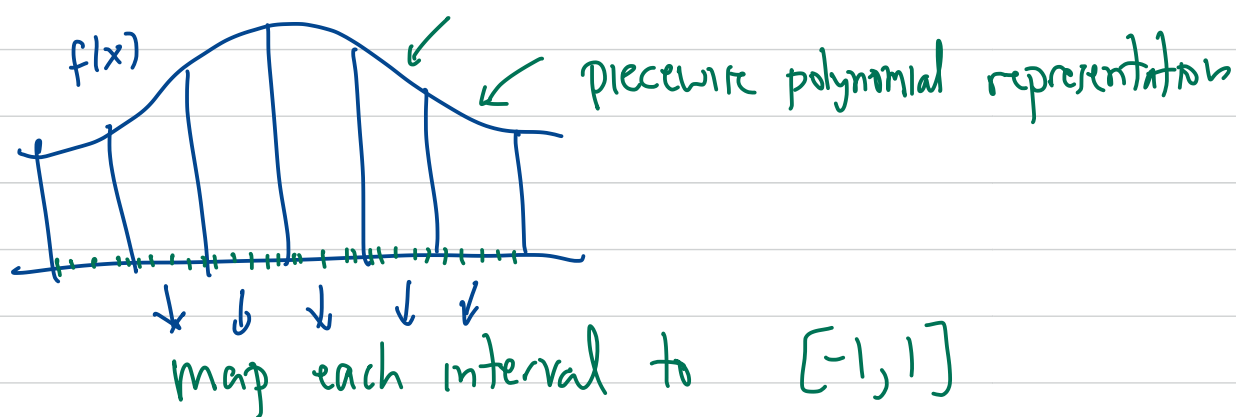
$$P_J f(x) = \frac{1}{2^{-J}} \int_{k/2^J}^{(k+1)/2^J} f(x) dx$$

for $\frac{k}{2^J} < x < \frac{k+1}{2^J}$ average value

each successive level adds corrections to the previous level
rather than storing the step function values directly

(TV video glitches yield blocky images rather than a
high-res sliver over a small fraction of the screen)

Chebyshev interpolation retains smoothness but gives up
self-orthogonality after rescaling for locality,



Write
$$f(x) = \sum_{j=0}^{\infty} a_n T_j(x) \quad -1 \leq x \leq 1$$

$$T_0(x) = 1, T_1(x) = x, T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$$

Theory closely parallels Fourier series

due to
$$T_n(\cos \theta) = \cos n\theta$$

(secretly a cosine expansion of $f(\cos \theta)$)